

DC014*Computer Science 1**Semester I**Session 2023/2024**2 hours***DC014****Sains Komputer 1****Semester I****Sesi 2023/2024****2 jam**

No. Matrik <i>Matric No.</i>									

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(Isikan maklumat dengan lengkap)**KEMENTERIAN PENDIDIKAN MALAYSIA****KOLEJ MATRIKULASI NEGERI SEMBILAN***NEGERI SEMBILAN MATRICULATION COLLEGE***PRA PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI***PRE MATRICULATION PROGRAMME SEMESTER EXAMINATION***JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.***DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.***D
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Untuk Kegunaan Pemeriksa <i>For Examiner's Use</i>			
No. Soalan Question No.	Markah Mark		
	Pemeriksa <i>Examiner</i>	KP / KKP	Markah Marks
1			6
2			16
3			6
4			4
5			18
6			20
JUMLAH TOTAL			70

Kertas soalan ini mengandungi **6** halaman bercetak.*This question paper consists of 6 printed pages.*

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1. Hannah and her team were asked by his boss to develop 4 Dimensional (4D) **Lets' Learn ABC** courseware for pre-school students in Malaysia. This courseware must be easy to be learn and can attract students' attention.

(a) Describe **one (1) most attractive element** that could be include in the courseware.

[2 marks]

(b) Explain the purpose of **most attractive element** mentioned in **1(a)**.

[2 marks]

(c) Differentiate between 3 Dimensional (3D) and 4 Dimensional (4D) representation of multimedia application.

[2 marks]

3 Dimensional (3D)	4 Dimensional (4D)

2. A multimedia product is made up of many ingredients from existing print products or from a multitude of authoring tools. Each product has its own set of requirements. There are three main phases to putting all elements in multimedia together.

(a) Describe **two (2) activities for each** of the first 2 phases in multimedia production.

[4 marks]

First Phase : _____	Second Phase : _____
Activity 1 :	Activity 1 :
Activity 2 :	Activity 2 :

(b) Differentiate among two types of multimedia authoring tools shown below.

[4 marks]

Characteristics	Time based	Icon based
Technique		
Arrangement of elements		

- (c) Explain **two (2)** types of multimedia navigation that allows user to obtain information in nonlinear way.

[4 marks]

Navigation Type 1 :	Navigation Type 2 :

- (d) Give **two (2)** differentiation between offline and online multimedia delivery.

[4 marks]

Offline	Online

3. A file system is a software that organizes and manages files on a storage media, whereas a database management system (DBMS) is a software that allows you to access, create, and administer databases.

- (a) Give **two (2)** differentiation between File Processing System and the Database System

[4 marks]

File Processing System	Database System

- (b) Describe **one (1)** disadvantage of using database system.

[2 marks]

4. A database object in a relational database is a data structure used to either store or reference data. The most common object that people interact with is **the table**. It is important to ensure allocated data in a system is safe, secure, and unaltered using data integrity.

- (a) Explain **one (1)** factor that determine the integrity of data stored in database system.

[2 marks]

(b) Explain the function **one (1)** of database object other than stated in question 4.

[2 marks]

5. Nowadays, even a small computers or devices can communicate directly with one another, with hundreds or thousands computer on a corporate network or with millions of others computer around the world via transmission media and communication devices.

(a) Describe commonly uses of networking application in the field below

(i) education

[2 marks]

(ii) healthcare

[2 marks]

(b) (i) Define the term of transmission media

[2 marks]

(ii) Explain **one (1)** example of wired transmission media

[2 marks]

(iii) Explain **one (1)** example wireless transmission media

[2 marks]

(c) (i) Define the term of communication device

[2 marks]

(ii) Describe the function of communication devices below.

[2 marks]

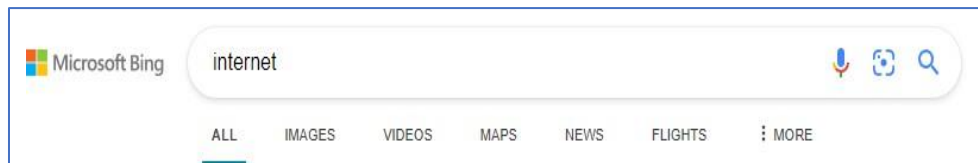
Communication Device	Function
Network Interface Card	
Wireless Access Point	

- (iii) Mr Ahmad, as a technician of Telekom Malaysia was instructed to connect UNIFI to Mr Rahman's house.

Suggest and explain **two (2)** devices that should be installed by Mr. Ahmad to ensure the effectiveness of the connection.

[4 marks]

6. The internet is widely used and readily accessible to hundreds of millions of people. Below is the diagram of search tool used to find instant information.



- (a) (i) State type of search tool shown on diagram.

[1 mark]

- (ii) Differentiate **two (2)** types of search tool.

[4 marks]

Types	Search tool X	Search tool Y
Description		

- (iii) Give **one (1)** example of search tool X.

[1 mark]

- (b) Searching the web more effectively doesn't really take much effort, but there are three search strategies you can use to search better and get more relevant results.

- (i) List **two (2)** search strategies that you can use for better relevant web results.

[2 marks]

- (ii) Describe **one (1)** of search strategies that you answer briefly.

[2 marks]

- (c) Internet is mainly used as a means of communication and for looking up information.
- (i) Based on web activities, explain the use of Internet Messaging and online entertainment.

[4 marks]

Web activities	
Internet Messaging	Online Entertainment

- (d) Electronic mail or email used to send and receive computer-stored messages via the internet. Many people mistakenly believe that the e-mail they send and receive is private and will never be read by anyone other than the intended recipient.
- (e) Describe **two (2)** tips how to improve your email's privacy.

[4 marks]

- (g) Explain **one (1)** personal application of Internet of Things (IoT).

[2 marks]

END OF QUESTION PAPER